

PROJECT 02

Active Directory Forest Deployment

Windows Server 2025 | AD DS | DNS | Domain Services

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ENVIRONMENT	Personal Windows Enterprise Lab
VERSION	v1.0.0
RELEASE DATE	August 5, 2026
STATUS	Complete

Executive Summary

Deployed the first Active Directory domain controller for the Windows enterprise lab. The project prepared **CORP-DC1** with static IPv4 and DNS settings, installed Active Directory Domain Services and DNS, promoted the server into a new **corp.mslab.test** forest, and validated directory, name-resolution, authentication, and policy-distribution services.

Business Problem

Windows clients, member servers, administrators, and enterprise applications require a centralized identity and naming platform. A carefully deployed first domain controller establishes the forest root, security boundary, DNS namespace, authentication services, and replication-ready directory structure that later labs will use for organizational units, delegated administration, Group Policy, DHCP, file services, and domain joins.

Objectives

- Create a rollback checkpoint before installing domain services.
- Configure CORP-DC1 with 172.16.10.10/24 and use itself as the preferred DNS server.
- Install Active Directory Domain Services and the DNS Server role.
- Create the corp.mslab.test forest with the CORP NetBIOS name.
- Use Windows Server 2025 forest and domain functional levels.
- Validate AD DS, DNS, KDC, Netlogon, NTDS, SYSVOL, NETLOGON, and domain-name resolution.

Environment

Component	Configuration
Virtualization	Microsoft Hyper-V; Generation 2 virtual machine
Domain controller	CORP-DC1 - Windows Server 2025 Standard Evaluation
Network	LAB-Corp-1 - 172.16.10.0/24 Internal switch
Server address	172.16.10.10/24; no default gateway
DNS client	Preferred DNS: 172.16.10.10
Forest / domain	corp.mslab.test; NetBIOS name CORP
Functional levels	Windows Server 2025 forest and domain
Scope boundary	No OUs, users, groups, DHCP, domain joins, CA, or Group Policy design

Architecture

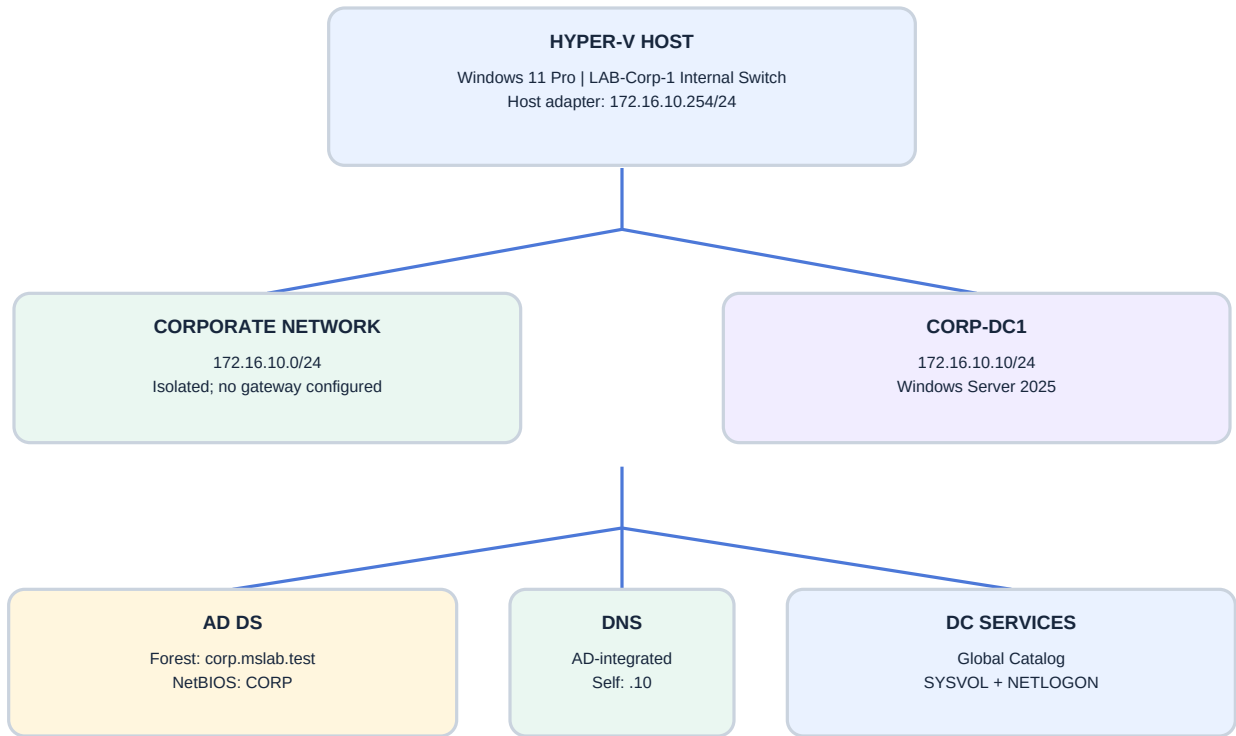


Figure 1 - Hyper-V network, CORP-DC1, and deployed directory services.

Directory Services Configuration

Setting	Value	Purpose
Forest root	corp.mslab.test	Creates the enterprise directory namespace
NetBIOS name	CORP	Supports legacy and short-form domain references
Domain controller	CORP-DC1	First writable DC and Global Catalog
AD DS database	C:\Windows\NTDS	Default directory database and log location
SYSVOL	C:\Windows\SYSVOL	Stores domain policies and logon scripts
DNS	AD-integrated on CORP-DC1	Provides domain-controller discovery and host resolution
Functional levels	Windows Server 2025	Uses the highest available lab capability level

Service Dependencies

Service	Role in the Domain
DNS	Locates the domain and domain-controller service records
KDC	Issues Kerberos tickets for domain authentication
Netlogon	Registers DC records and supports secure-channel operations
NTDS	Hosts Active Directory Domain Services
SYSVOL / NETLOGON	Distributes policy and script content to domain members

Implementation Summary

1. Created and renamed a pre-deployment Hyper-V checkpoint as p02-PreADDS.
2. Confirmed the server name CORP-DC1 before domain-controller promotion.
3. Assigned 172.16.10.10/24, left the gateway blank for lab isolation, and configured 172.16.10.10 as preferred DNS.
4. Installed the Active Directory Domain Services and DNS Server roles.
5. Selected Add a new forest and defined corp.mslab.test as the root domain.
6. Confirmed the CORP NetBIOS name and Windows Server 2025 forest and domain functional levels.
7. Enabled DNS Server and Global Catalog, kept the domain controller writable, and retained the default NTDS and SYSVOL paths.
8. Accepted the expected DNS-delegation warning because no authoritative parent zone exists in the isolated lab.
9. Passed the AD DS prerequisite check, promoted CORP-DC1, and allowed the server to restart.
10. Validated domain, forest, services, shares, DNS resolution, and DCDiag test results.

Configuration Standards

Control	Applied Configuration	Operational Value
Pre-change recovery	p02-PreADDS checkpoint	Creates a rollback baseline before role installation
Addressing	Static 172.16.10.10/24	Prevents DNS and domain services from changing addresses
DNS client	Points to 172.16.10.10	Ensures the DC queries its AD-integrated DNS service
Global Catalog	Enabled	Provides forest-wide object lookup and logon support
RODC	Disabled	Deploys the first domain controller as writable
Data paths	Default NTDS and SYSVOL	Appropriate for this single-disk training environment

SCOPE

Project 02 stops after forest deployment and service validation. Organizational units, users, groups, domain joins, DHCP, Group Policy, certificate services, and additional domain controllers remain separate labs.

Evidence - Pre-Deployment Recovery and Static Networking

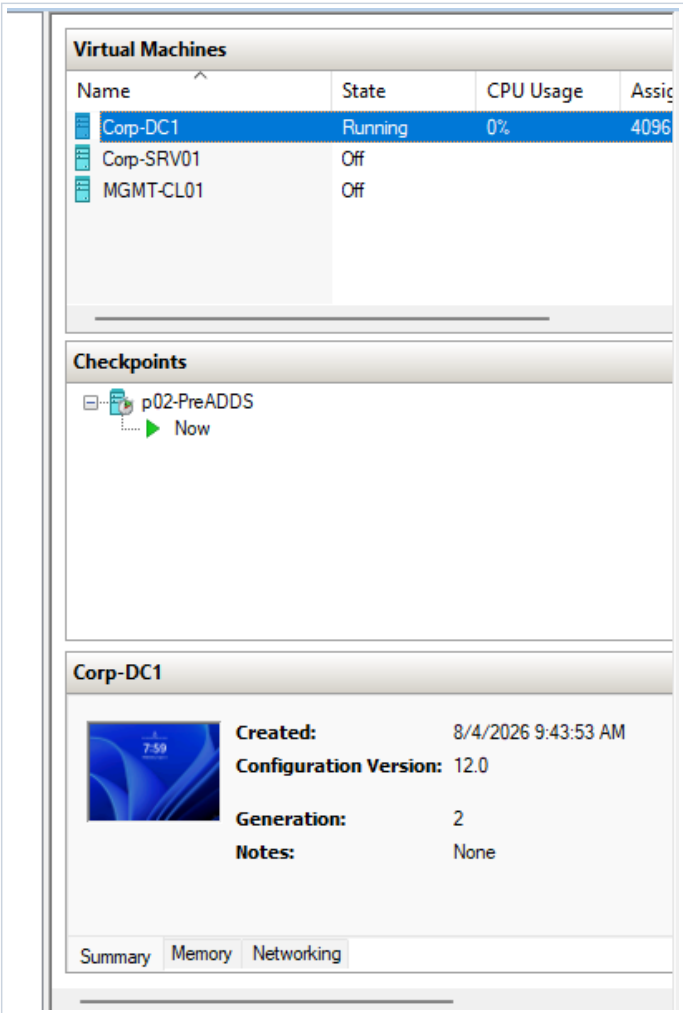


Figure 2 - Pre-AD DS Hyper-V Checkpoint

Hyper-V Manager shows p02-PreADDS created before directory-service installation, providing a controlled rollback point.

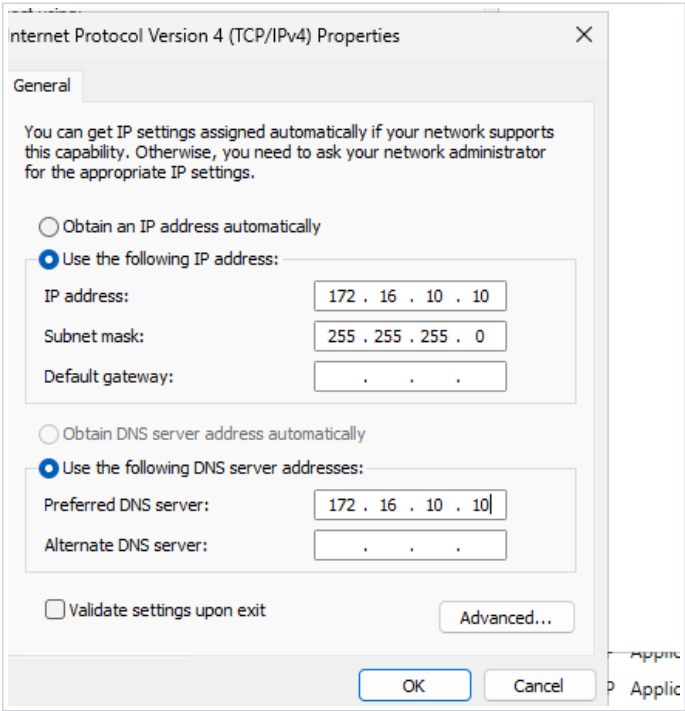


Figure 3 - CORP-DC1 Static IPv4 and DNS Configuration

CORP-DC1 uses 172.16.10.10/24 with no gateway and points preferred DNS to its own planned DNS service at 172.16.10.10.

Why This Baseline Matters

A domain controller must retain stable network identity. Static addressing prevents clients and service records from referencing a changing address, while self-referential DNS enables the first DC to locate the new AD-integrated namespace after promotion. The absent gateway is intentional because routing has not yet been introduced to the isolated LAB-Corp-1 network.

RECOVERY

The checkpoint is a pre-deployment safety control for this lab stage. Once other systems depend on the domain, routine rollback of an old domain-controller snapshot should be avoided in favor of supported backup and recovery procedures.

Evidence - AD DS, DNS, and Forest Configuration

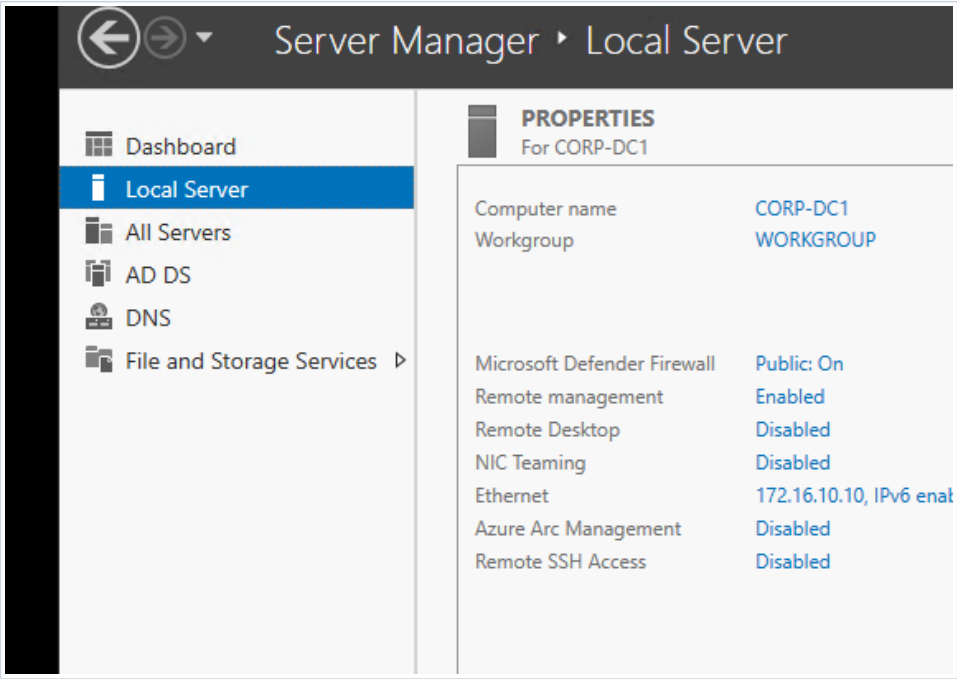


Figure 4 - Active Directory Domain Services and DNS Installed
Server Manager displays the AD DS and DNS management nodes after the required roles were installed on CORP-DC1.

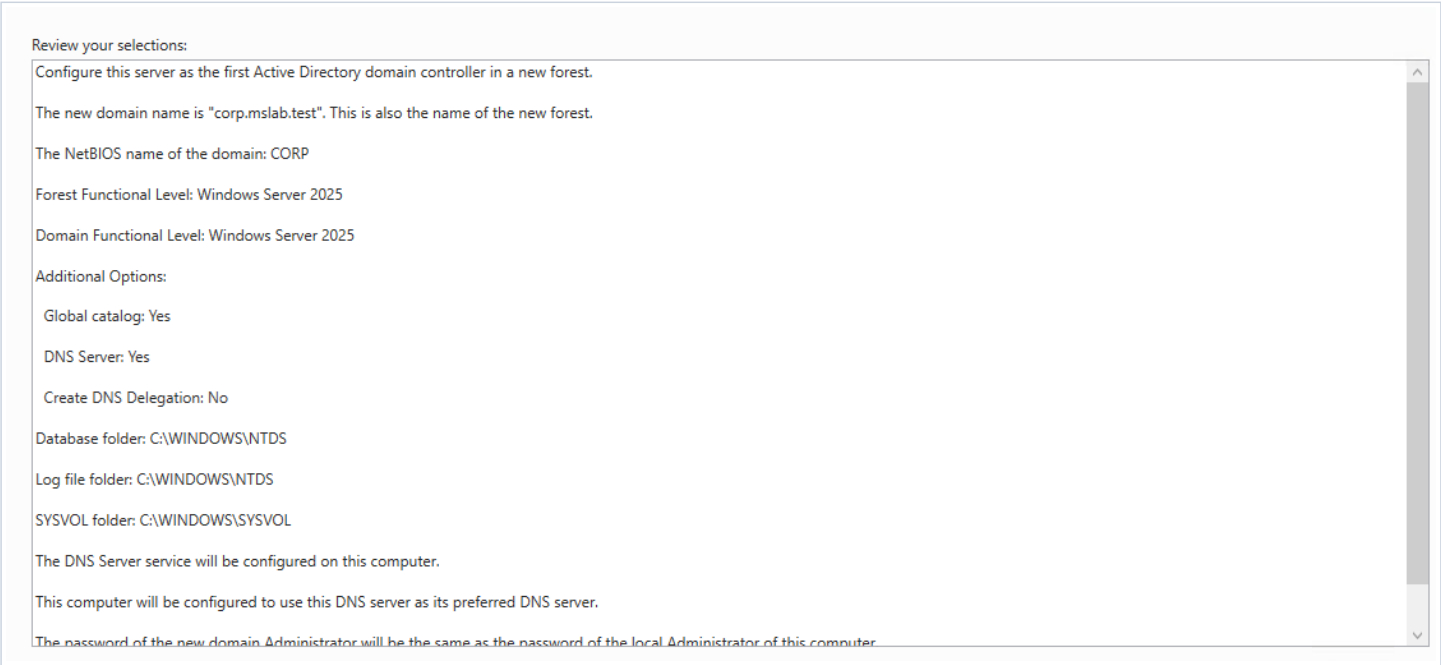


Figure 5 - New Forest Promotion Settings
The review screen records corp.mslab.test, CORP, Windows Server 2025 functional levels, Global Catalog and DNS enabled, and default NTDS and SYSVOL paths.

Evidence - Domain Controller Promotion Readiness

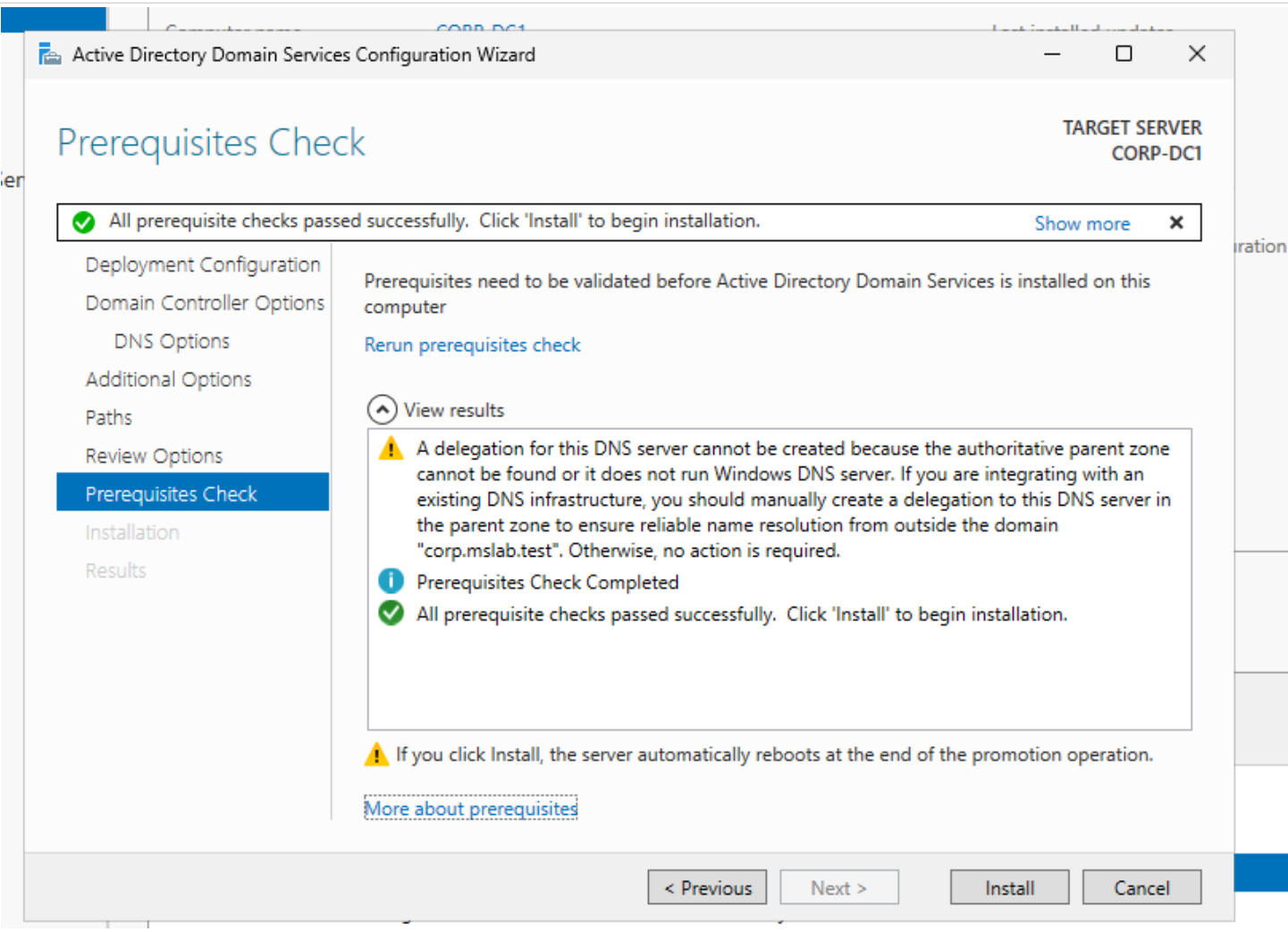


Figure 6 - AD DS Prerequisite Check Passed

The promotion wizard reports that all prerequisite checks passed successfully. The DNS delegation warning is expected for a new isolated forest without an authoritative parent zone.

Promotion Readiness Review

Check	Result	Interpretation
Required prerequisite checks	PASS	Wizard approved installation
New forest root	PASS	corp.mslab.test
Target server	PASS	CORP-DC1
DNS delegation	EXPECTED WARNING	No parent DNS zone exists in this isolated lab
Automatic restart	EXPECTED	Promotion restarts the server to activate AD DS

DECISION

The DNS delegation message did not block deployment because corp.mslab.test is the root of a new forest, not a delegated child of an existing authoritative DNS namespace.

Evidence - Active Directory and DNS Health Validation

```
DNSRoot          NetBIOSName          DomainMode
-----          -
corp.mslab.test  CORP              Windows2025Domain

Directory Server Diagnosis

Performing initial setup:
  Trying to find home server...
  Home Server = CORP-DC1
  * Identified AD Forest.
  Done gathering initial info.

Doing initial required tests

  Testing server: Default-First-Site-Name\CORP-DC1
    Starting test: Connectivity
      ..... CORP-DC1 passed test Connectivity

Doing primary tests

  Testing server: Default-First-Site-Name\CORP-DC1
    Starting test: Advertising
      ..... CORP-DC1 passed test Advertising
    Starting test: SysVolCheck
      ..... CORP-DC1 passed test SysVolCheck
    Starting test: Services
      ..... CORP-DC1 passed test Services

    Starting test: DNS

      DNS Tests are running and not hung. Please wait a few minutes...
      ..... CORP-DC1 passed test DNS

Running partition tests on : ForestDnsZones

Running partition tests on : DomainDnsZones

Running partition tests on : Schema

Running partition tests on : Configuration

Running partition tests on : corp

Running enterprise tests on : corp.mslab.test
  Starting test: DNS
    ..... corp.mslab.test passed test DNS
```

Figure 7 - DCDiag Domain Controller Validation

DCDiag confirms Connectivity, Advertising, SysVolCheck, Services, and DNS tests passed for CORP-DC1 and corp.mslab.test.

Evidence - Forest, Services, Shares, and Name Resolution

```
RootDomain          ForestMode
-----
corp.mslab.test Windows2025Forest

PS C:\Users\Administrator> Get-Service NTDS,DNS,Netlogon,KDC |
    Format-Table Name,Status,StartType -AutoSize

Name      Status StartType
----      -
DNS       Running Automatic
KDC       Running Automatic
Netlogon  Running Automatic
NTDS      Running Automatic

PS C:\Users\Administrator> Get-SmbShare -Name SYSVOL,NETLOGON |
    Format-Table Name,Path -AutoSize

Name      Path
----      -
NETLOGON  C:\WINDOWS\SYSVOL\sysvol\corp.mslab.test\SCRIPTS
SYSVOL    C:\WINDOWS\SYSVOL\sysvol

PS C:\Users\Administrator> Resolve-DnsName corp.mslab.test -Type A
Resolve-DnsName CORP-DC1.corp.mslab.test -Type A

Name                                     Type  TTL  Section  IPAddress
----
corp.mslab.test                         A      600  Answer   172.16.10.10
CORP-DC1.corp.mslab.test                A     1200  Question 172.16.10.10

PS C:\Users\Administrator>
```

Figure 8 - Post-Promotion PowerShell Validation

The output confirms Windows2025Forest, running automatic DNS/KDC/Netlogon/NTDS services, SYSVOL and NETLOGON shares, and A-record resolution to 172.16.10.10.

Validated Outcomes

Validation Area	Confirmed Result
Forest root / mode	corp.mslab.test / Windows2025Forest
Core services	DNS, KDC, Netlogon, and NTDS running automatically
Domain shares	SYSVOL and NETLOGON available
Domain A record	corp.mslab.test -> 172.16.10.10
DC host A record	CORP-DC1.corp.mslab.test -> 172.16.10.10

Validation Results

Validation	Result	Evidence
Pre-AD DS checkpoint created	PASS	Figure 2
Static IP and preferred DNS configured	PASS	Figure 3
AD DS and DNS roles installed	PASS	Figure 4
Forest, NetBIOS name, levels, GC, DNS, and paths reviewed	PASS	Figure 5
Promotion prerequisites completed	PASS	Figure 6
Connectivity, Advertising, SYSVOL, Services, and DNS health	PASS	Figure 7
Forest mode, services, shares, and A records	PASS	Figure 8

Security Analysis

The forest root is a high-value security boundary because it establishes the identities, authentication services, administrative roles, and policy-distribution mechanisms for the entire lab. CORP-DC1 therefore uses a stable address, isolated virtual network, writable first-DC configuration, DNS-integrated namespace, and validated Kerberos, Netlogon, directory, SYSVOL, and DNS services.

The pre-change checkpoint improves recoverability during initial deployment, but it is not a substitute for system-state backup, additional domain controllers, monitored replication, privileged-access separation, or tested forest recovery. Those controls become more important as member systems and administrative accounts are added.

Evidence Limitations

The supplied evidence confirms the checkpoint, network settings, installed server roles, forest configuration, successful prerequisite check, DCDiag results, core service state, domain shares, and DNS resolution. A separate screenshot of Active Directory Users and Computers, DNS Manager zones and SRV records, Event Viewer, Windows Firewall profile, or a system-state backup was not included in this edition. Those omissions do not contradict the validated deployment, but they identify useful evidence for later operational-hardening labs.

Lessons Learned

- The first domain controller should have a stable server name and IPv4 address before promotion.
- Active Directory depends on DNS; domain clients must use the AD-aware DNS service rather than an unrelated resolver.
- A DNS delegation warning is expected when creating the root of a new isolated forest without a parent zone.
- SYSVOL and NETLOGON are operational checks, not optional artifacts; both support domain policy and logon behavior.
- DCDiag, service-state checks, share validation, and DNS resolution provide stronger evidence than relying only on a successful wizard.

Production Improvements

- Deploy at least one additional domain controller and verify AD DS and DNS replication.
- Configure supported system-state backups and perform documented restore testing.
- Use dedicated privileged accounts and a hardened administrative workstation for domain administration.
- Configure DNS forwarders, time synchronization, monitoring, alerting, and centralized event collection.
- Apply security baselines, Windows Firewall policy, patch management, and audited administrative delegation.
- Place AD DS database, logs, and SYSVOL according to production capacity, resilience, and recovery requirements.

Skills Demonstrated

Active Directory	Networking / DNS	Validation / Operations
New forest deployment	Static IPv4 configuration	DCDiag health testing
Domain-controller promotion	AD-integrated DNS	Windows service validation
Forest and domain levels	A-record resolution	SYSVOL and NETLOGON checks
Global Catalog configuration	Isolated subnet design	Checkpoint-based change control

Resume Bullet

RESUME	Deployed and validated a Windows Server 2025 Active Directory forest on Hyper-V, configuring static IPv4 and AD-integrated DNS, promoting CORP-DC1 for corp.mslab.test, and verifying Kerberos, Netlogon, NTDS, SYSVOL, NETLOGON, and DNS health with PowerShell and DCDiag.
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STAR Interview Story

Situation: The Hyper-V foundation was complete, but the lab had no centralized identity, authentication, or domain-aware name-resolution service.

Task: Deploy the first domain controller and establish a reliable forest root that could safely support future users, servers, clients, policies, and enterprise services.

Action: I created a pre-change checkpoint, assigned CORP-DC1 a static address, configured it to use its planned DNS service, installed AD DS and DNS, created the corp.mslab.test forest at Windows Server 2025 functional levels, and validated the deployment with DCDiag, PowerShell, service checks, share checks, and DNS queries.

Result: CORP-DC1 now provides a documented and validated identity and DNS foundation with passing core health tests, ready for directory design and controlled onboarding of the management client and member server.

Next Lab Handoff

Project 03 will design the Active Directory organizational-unit structure and administrative model for corp.mslab.test. It will establish logical containers for users, groups, computers, servers, workstations, administrators, and service accounts before CORP-SRV01 and MGMT-CL01 are joined to the domain.

Version History

Version	Release Date	Description
v1.0.0	August 5, 2026	Initial documented implementation. Configured CORP-DC1 networking, installed AD DS and DNS, created the corp.mslab.test forest, completed domain-controller promotion, validated directory and DNS health, and embedded all supplied evidence using portfolio documentation standard v1.0.1.

References

- Microsoft Learn - Install Active Directory Domain Services
- Microsoft Learn - Install and configure DNS Server on Windows Server
- Microsoft Learn - DCDiag command-line reference
- Microsoft Learn - Active Directory Domain Services overview
- User-provided Hyper-V checkpoint, network, Server Manager, forest-promotion, prerequisite, DCDiag, service, share, and DNS evidence